Project Design Proposal

Predicting Voluntary Turnover Using Machine Learning Classification Methods for Strategic Retention Planning

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Problem Heading and Context

Business Problem: Which employees are most likely to leave, and why?

Voluntary employee turnover remains one of the most persistent and costly challenges facing modern organizations. Replacing an employee can cost up to 213% of their annual salary, depending on the job and type ok skill (Boushey & Glynn, 2012). Beyond direct financial costs, turnover disrupts team cohesion and reduces productivity. As organizations increasingly rely on knowledge workers and specialized talent, the ability to anticipate and prevent voluntary exits has become a strategic priority.

Traditional Human Resources (HR) approaches to turnover—such as exit interviews, engagement surveys, and manager intuition—tend to be reactive, subjective, and limited in predictive power. They explain why employees left but do not effectively identify who is likely to leave next or what factors are most influential in advance. In contrast, machine learning (ML) enables organizations to analyze large volumes of employee data, uncover hidden patterns, and generate early-warning indicators of attrition risk. This aligns directly with the project theme: using ML classification methods to predict voluntary turnover and support strategic retention planning.

Employee turnover imposes substantial financial and operational burdens on organizations, making early identification of at-risk employees a critical HR priority. Research shows that machine-learning (ML) models—including logistic regression, random forests, gradient boosting, and neural networks—consistently outperform traditional statistical approaches in predicting employee attrition (Guerranti & Dimitri, 2023; Regasse & Venier, 2025). These models capture nonlinear relationships and complex interactions that conventional methods often fail to detect.

Beyond predictive accuracy, scholars emphasize the importance of interpretability to ensure transparency, fairness, and trust in HR decision-making. Explainability frameworks such as SHAP and LIME have become widely adopted for interpreting ML predictions, enabling practitioners to understand how features such as job satisfaction, tenure, and overtime contribute to turnover risk (Lundberg & Lee, 2017; Ribeiro et al., 2016). Recent organizational research highlights that transparent and interpretable AI systems are essential for ethical and responsible HR analytics, particularly in sensitive domains such as hiring and turnover prediction (Van den Broek et al., 2021).

Building on this foundation, the proposed project aims to develop a predictive model that identifies employees at risk of voluntary turnover and explains the key drivers behind these predictions. This approach supports evidence-based workforce planning, targeted retention strategies, and more transparent HR decision-making.

Research Questions and Justifications

This study will address the following four interconnected research questions:

- **RQ1 - Which employees are most likely to voluntarily leave the organization?**
  This question directly addresses the core business problem and aligns with the theme of predictive classification. Identifying high-risk employees enables proactive retention interventions, reducing turnover costs and operational disruption.

- **RQ2 - Which employee characteristics are most predictive of voluntary turnover?**
  Understanding the most influential factors (e.g., job satisfaction, compensation, tenure, workload) allows organizations to focus retention resources where they have the greatest impact. This supports evidence-based HR decision-making.

- **RQ3 - Which machine learning classification methods provide the most accurate and reliable predictions of voluntary turnover?**
  Comparing multiple ML algorithms (e.g., Logistic Regression, Random Forest, XGBoost) ensures methodological rigor and helps identify the best-performing model for HR decision-making. It also demonstrates whether advanced ML techniques offer meaningful improvements over simpler models such as Logistic Regression. This question also supports academic expectations for experimentation and evaluation.

- **RQ4 - How can model outputs be translated into practical retention strategies for workforce planning?**
  The goal is not only prediction but also strategic intervention. Linking predictions to HR actions increases business value. This bridges technical modeling with real-world business impact, demonstrating how predictive analytics can inform HR policy, resource allocation, and long-term workforce planning.

Data Selection and Rationale

Dataset Description

The project will use the IBM HR Analytics Employee Attrition Dataset (IBM Watson Analytics, 2016), a widely recognized and publicly available dataset with similar variables used in academic research on turnover prediction (Saradhi & Palshikar, 2011). The dataset contains 1,470 employee records with 35 employee-level features variables including:

- Age
- Gender

- Job Role
- Monthly Income
- Job Satisfaction
- Work-Life Balance
- Overtime
- Years at Company
- Performance Rating
- Distance from Home
- Attrition (Yes/No – target variable)

Dataset Justification

- Relevance: Contains the variables typically used in real-world turnover modeling and commonly available in HR systems.
- Feasibility: Clean, structured, and ready for ML pipelines without requiring sensitive organizational data.
- Richness and Realism: Includes both behavioral and organizational features, enabling robust feature engineering.
- Benchmarking: Extensively used in prior studies, allowing comparison with established results.
- Ethical and practical: Avoids privacy concerns associated with real HRIS data.
- Analytical Depth: The dataset allows for feature importance analysis, model comparison, and interpretability techniques (e.g., SHAP values).

Potential Limitations

The IBM HR Analytics Employee Attrition dataset carries several important limitations that should be acknowledged in any academic analysis. Because it is a synthetic dataset designed to mimic a generic corporate environment, it may not fully capture the complexity or nuance of real organizational dynamics. Its lack of temporal depth also prevents the use of time-series or survival analysis techniques, which are often valuable in turnover research. Additionally, the dataset does not differentiate between voluntary and involuntary attrition, limiting the precision of predictive modeling. Several variables, such as job satisfaction, are self-reported and therefore subject to bias, reducing their reliability as predictors.

Project will focus on methodological rigor and interpretability rather than organizational deployment.

Proposed Methodology and Tools

Research Methodology Framework

To address the research questions effectively, the project will follow the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework. This iterative approach ensures that the technical models remain anchored to the business problem of employee retention.

Phase I: Data Preprocessing & Exploratory Data Analysis (EDA)
Before modeling, the IBM dataset will undergo a structured preparation pipeline to ensure the data is clean, balanced, and optimized for machine learning algorithms.

- **Step 1: Data Cleaning & Feature Engineering**: review of missing values and outliers. Categorical variables (e.g., Job Role, Marital Status) will be converted into numerical formats using One-Hot Encoding to make the data machine-readable.
- **Step 2: Correlation Analysis & EDA:** Using Heatmaps to identify multi-collinearity between variables such as "Years at Company" and "Years in Current Role." This allows for early identification of redundant features before they bias the model.
- **Step 3: Feature Scaling:** Applying Standard or Robust Scaling to numerical features (e.g., Monthly Income, Age). This ensures that features with larger ranges do not disproportionately influence the model's weight distribution.
- **Step 4: Handling Class Imbalance:** Since attrition datasets are typically skewed (fewer "Left" than "Stayed" employees for example), we will implement SMOTE (Synthetic Minority Over-sampling Technique). This creates synthetic examples of the minority class to ensure the model learns the characteristics of turnover rather than just memorizing the majority.
- **Step 5: Dimensionality Reduction (PCA):** If the feature set remains high-dimensional after encoding, Principal Component Analysis (PCA) will be used to compress the features into a set of uncorrelated components, reducing noise and improving computational efficiency.

Phase II: Predictive Modeling (Classification)
To answer RQ1 and RQ3, the following comparative modeling strategy using three distinct tiers of algorithms will be implemented:

1. **Baseline Model:** *Logistic Regression* to provide a benchmark for interpretability.
2. **Ensemble Models:** *Random Forest* and *XGBoost* to capture the non-linear relationships and complex interactions.
3. **Optimization:** Hyperparameter Tuning via Grid Search or Random Search to maximize the performance of each model.

Phase III: Model Evaluation & Selection
Models will be evaluated not just on accuracy, but on metrics that reflect the "cost of turnover":

- **Recall (Sensitivity):** Crucial for HR, as it measures the ability to find *all* employees likely to leave (minimizing False Negatives).
- **F1-Score:** To balance precision and recall.
- **ROC-AUC:** To evaluate the model's ability to distinguish between the two classes.

Phase IV: Explainability & Strategy (Text/Feature Mining)
- To address RQ2 and RQ4, XAI (Explainable AI) techniques will be applied:

- **SHAP (SHapley Additive exPlanations):** To quantify exactly how much each feature (like "Overtime" or "Monthly Income") contributed to an individual's attrition score.
- **Feature Importance Ranking:** Identifying the top 5 global drivers of turnover to inform corporate policy.

Technical Tools & Technologies

The following tools have been selected based on their robustness in academic research and industry application:

Category	Tool/Library	Purpose
Programming Language	Python 3.x	The primary language for data science and ML pipelines.
Data Manipulation	Pandas & NumPy	For cleaning, filtering, and transforming the IBM dataset.
Machine Learning	Scikit-Learn	Implementation of Logistic Regression and Random Forest.
Advanced Boosting	XGBoost / LightGBM	For high-performance gradient boosting classification.
Interpretability	SHAP / LIME	To explain "Why" the model predicts an employee will leave.
Visualization	Matplotlib / Seaborn	For EDA and presenting findings to HR stakeholders.

Feasibility & Alignment

Research Question	Methodology Approach	Feasibility Consideration
RQ1: Who is leaving?	Binary Classification (Yes/No)	High; Target variable "Attrition" is clearly defined in the IBM dataset.
RQ2: What are the drivers?	Feature Importance & SHAP Values	High; Dataset contains 35 features covering diverse HR domains.
RQ3: Best model?	Comparative Performance Analysis	High; Standard ML metrics (Recall/AUC) are easy to compute and compare.
RQ4: Strategic Action?	Prescriptive Analytics	Moderate; Requires translating data "drivers" into HR policy recommendations.

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